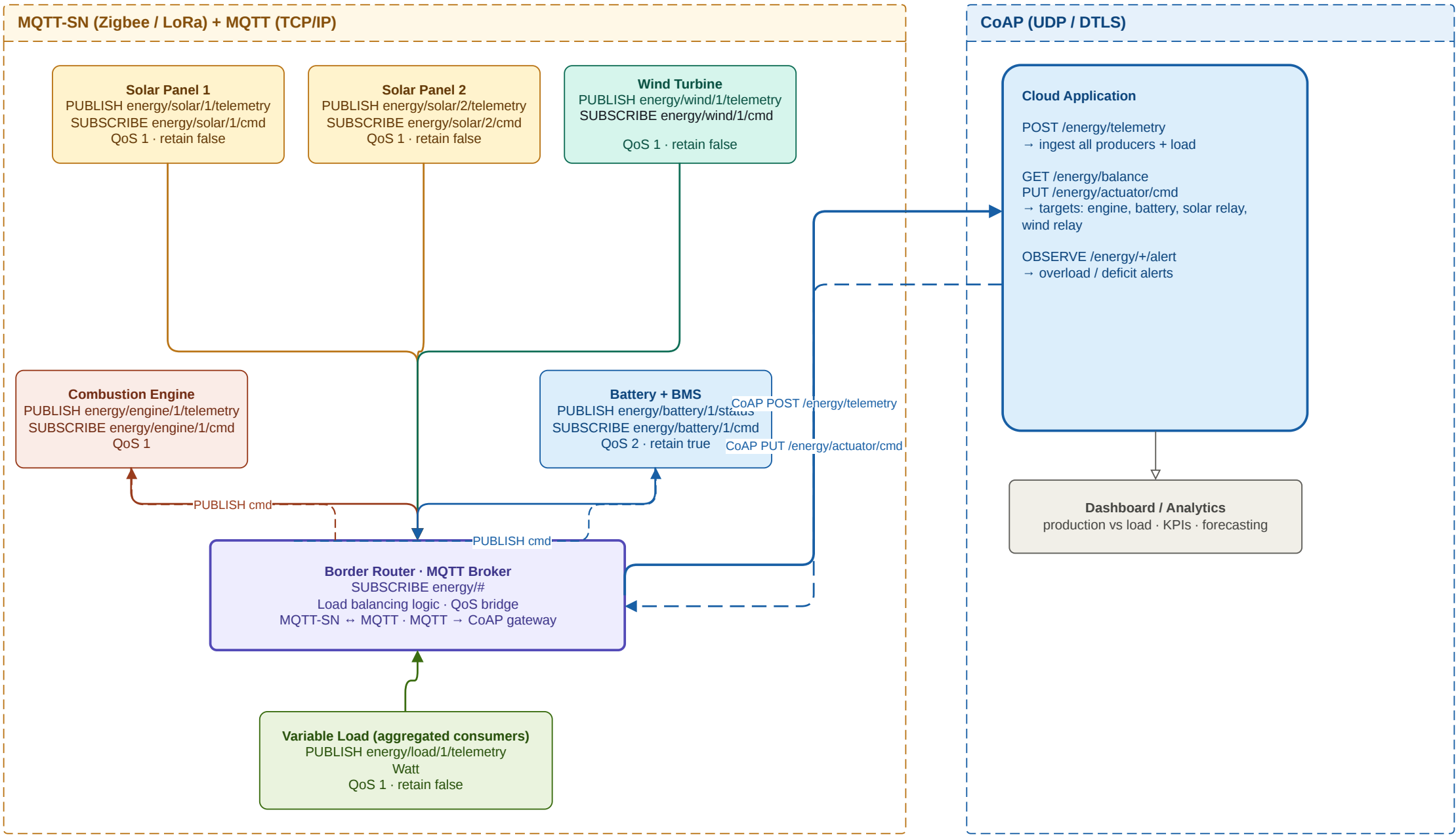




Payload examples

MQTT energy/solar/1/telemetry → {"W":1240, "V":48.2, "I":25.8, "irr":820, "t":38.1, "ts":1747734000}
MQTT energy/load/1/telemetry → {"W":3400, "ts":1747734000}
CoAP POST /energy/telemetry → CBOR {src:"load/1", W:3400, ts:1747734000}
CoAP PUT /energy/actuator/cmd → CBOR {target:"engine/1", action:"start", setpoint_W:2000}
MQTT energy/solar/1/cmd → {"relay":"off", "reason":"battery_full", "ts":...}

Legend

— MQTT PUBLISH telemetry (sensor data upstream)
----- MQTT PUBLISH cmd / CoAP PUT (commands downstream to actuators)
QoS 1 = at least once (telemetry) · QoS 2 = exactly once (battery cmd) · CBOR = compact binary payload